TRANSFORM CALCULUS, FOURIER SERIES AND NUMERICAL TECHNIQUES

(Course Code: 23MA3BSTFN)

UNIT 1 - LAPLACE TRANSFORMS

Laplace Transform of $f(t)$ denoted by $L(f(t))$ is defined by $L(f(t)) = \int_0^\infty e^{-st} f(t)\,dt = F(s)$ provided that the integral exists where s is a parameter which may be a real or complex number.

Transforms of Elementary functions

1. $L(k) = \dfrac{k}{s}$, $s > 0$, where k is a constant
2. $L(t^n) = \begin{cases} \dfrac{n!}{s^{n+1}} & \text{when n is a positive integer} \\ \dfrac{\Gamma(n+1)}{s^{n+1}} & \text{for n>-1 and s>0} \end{cases}$
3. $L(e^{+at}) = \dfrac{1}{s-a}$, $s > a$ and $L(e^{-at}) = \dfrac{1}{s+a}$, $s > -a$
4. $L(\sin at) = \dfrac{a}{s^2 + a^2}$
5. $L(\cos at) = \dfrac{s}{s^2 + a^2}$
6. $L(\cosh at) = \dfrac{s}{s^2 - a^2}$
7. $L(\sinh at) = \dfrac{a}{s^2 - a^2}$

Properties of Laplace transforms (Statement only)

1. **Linearity property**: If $L(f(t)) = F(s)$, $L(g(t)) = G(s)$ and $L(h(t)) = H(s)$ and let *a, b, c* be any constants then $L(af(t) + bg(t) - ch(t)) = aF(s) + bG(s) - cH(s)$.
2. **Shifting property** : If $L(f(t)) = F(s)$, then $L\left(e^{at} f(t)\right) = F(s-a)$
3. **Multiplication by t** : If $L(f(t)) = F(s)$, then $L(t\,f(t)) = -\dfrac{d}{ds}(F(s))$.

   In general $L(t^n f(t)) = (-1)^n \dfrac{d^n}{ds^n}(F(s))$ *for* $n = 1, 2, 3, ...$
4. **Division by t** : If $L(f(t)) = F(s)$, then $L\left(\dfrac{f(t)}{t}\right) = \int_s^\infty F(s)\,ds$.
5. **Laplace transform of derivatives:** If $L(f(t)) = F(s)$, then $L(f'(t)) = sF(s) - f(0)$.

   In general $L(f^n(t)) = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - ..... - f^{n-1}(0)$

6. **Laplace transform of integral:** If $L\left(f(t)\right)=F(s)$, then

$$L\left(\int_0^t f(u)\,du\right)=\frac{F(s)}{s}, \quad s>0.$$

Find the Laplace transform for the following function $f(t)$ using the definition.

1. $f(t)=\begin{cases}\cos(t-2\pi/3) & t>2\pi/3\\ 0 & t<2\pi/3\end{cases}$ $\qquad Ans:\ e^{-2\pi s/3}\left[\dfrac{s}{s^2+1}\right]$

2. $f(t)=\begin{cases}t^2 & 0<t<2\\ t-1 & 2<t<3\\ 7 & t>3\end{cases}$

   $Ans:\ \dfrac{2}{s^3}-\dfrac{e^{-2s}}{s^3}\left(2+3s+3s^2\right)+\dfrac{e^{-3s}}{s^2}\left(5s-1\right)$

3. $f(t)=\begin{cases}\sin 2t & 0<t<\pi\\ 0 & t>\pi\end{cases}$

4. $f(t)=\begin{cases}e^t & 0<t<1\\ 0 & t>1\end{cases}$.

Find the Laplace transform for the following function $f(t)$ using linearity property.

1. $f(t)=\left(\sqrt{t}\pm\dfrac{1}{\sqrt{t}}\right)^3$ $\qquad$ Ans: $\dfrac{\sqrt{\pi}}{4}\left[\dfrac{3}{s^{5/2}}\pm\dfrac{6}{s^{3/2}}+\dfrac{12}{s^{1/2}}\pm\dfrac{8}{s^{-1/2}}\right]$

2. $f(t)=4e^{5t}+6t^3-3\sin 4t+2\cos 2t$ $\qquad$ Ans : $\dfrac{4}{s-5}+\dfrac{36}{s^4}-\dfrac{12}{s^2+16}+\dfrac{2s}{s^2+4}$

3. $f(t)=\cos 3t\cos 2t\cos t$ $\qquad$ Ans : $\dfrac{1}{4}\left(\dfrac{s}{s^2+32}+\dfrac{s}{s^2+16}+\dfrac{s}{s^2+4}+\dfrac{1}{s}\right)$

4. $f(t)=3\sinh 5t-4\cosh 5t$ $\qquad$ Ans: $\dfrac{3-4s}{s^2-25}$.

Find the Laplace transform for the following function $f(t)$ using shifting property.

1. $f(t)=(t+2)^2e^t$ $\qquad$ Ans: $\dfrac{4s^2-4s+2}{\left(s-1\right)^3}$

2. $f(t)=e^{4t}\sin 2t\cos t$ $\qquad$ Ans: $\dfrac{1}{2}\left(\dfrac{3}{\left(s-4\right)^2+9}+\dfrac{1}{\left(s-4\right)^2+1}\right)$

3. $f(t)=e^{-4t}t^{-5/2}$ $\qquad$ Ans:

4. $f(t)=(1+te^{-t})^3$ $\qquad$ Ans: $\dfrac{1}{s}+\dfrac{3}{(s+1)^2}+\dfrac{6}{(s+2)^3}+\dfrac{6}{(s+3)^4}$

5. $f(t)=e^{-4t}\cosh 2t$ $\qquad$ Ans: $\dfrac{s+4}{s^2+8s+12}$.

6. $f(t)=\sinh at\sin at$ $\qquad$ Ans: $\dfrac{2a^2s}{\left(s^4+4a^4\right)}$.

Find the Laplace transform for the following function using Multiplication by t property.

1. $f(t) = t\sin^2 t$ Ans: $\dfrac{2(3s^2+4)}{s^2\left(s^2+4\right)^2}$
2. $f(t) = \sin 2t - 2t\cos 2t$ Ans: $\dfrac{16}{\left(s^2+4\right)^2}$
3. $f(t) = te^{2t}\sin 3t$ Ans: $\dfrac{6(s-2)}{(s^2-4s+13)^2}$
4. $f(t) = t^2\sin t$

Find the Laplace transform for the following function using division by t property.

1. $f(t) = \dfrac{e^{-at}-e^{-bt}}{t}$ Ans: $\log\left(\dfrac{s+b}{s+a}\right)$
2. $f(t) = \dfrac{\sin^2 t}{t}$ Ans: $\dfrac{1}{4}\log\left(\dfrac{s^2+4}{s^2}\right)$
3. $f(t) = 2^t + \dfrac{\cos at - \cos bt}{t} + t\sin t$ Ans: $\dfrac{1}{s-\log 2} + \dfrac{1}{2}\log\left(\dfrac{s^2+b^2}{s^2+a^2}\right) + \dfrac{s^2-1}{\left(s^2+1\right)^2}$
4. $f(t) = \dfrac{\sin 3t\cos t}{t}$ Ans: $\dfrac{1}{2}\left[\pi - \tan^{-1}\left(\dfrac{s}{4}\right) - \tan^{-1}\left(\dfrac{s}{2}\right)\right]$
5. $f(t) = \dfrac{e^{-t}\sin t}{t}$ Ans: $\cot^{-1}(s+1)$

Evaluate the following using Laplace transform.

1. $L\left(e^{-t}\int_0^t \dfrac{\sin t}{t}dt\right)$ Ans: $\dfrac{\cot^{-1}(s+1)}{s+1}$
2. $L\left(t\int_0^t \dfrac{e^{-t}\sin t}{t}dt\right)$ Ans: $\dfrac{s+(s^2+2s+2)\cot^{-1}(s+1)}{s^2(s^2+2s+2)}$
3. $L\left(\int_0^t \dfrac{1-e^{-t}}{t}dt\right)$ Ans: $\dfrac{1}{s}\log\left(\dfrac{s+1}{s}\right)$
4. $L\left(\int_0^t\int_0^t\int_0^t t\sin t\,dt\,dt\,dt\right)$ Ans: $\dfrac{2}{s^2\left(s^2+1\right)^2}$

Periodic function: A function $f(t)$ is said to be periodic of period T if $f(t+T) = f(t)$ for $T > 0$.

Laplace transform of a periodic function.(statement only)

If $f(t)$ is a periodic function with period $T > 0$ then $L[f(t)] = \dfrac{1}{1-e^{-sT}}\int_0^T e^{-st}f(t)dt$

1. Find the Laplace transform of the square wave function of period a defined as

$f(t)=\begin{cases}1 & when\ \ 0<t<a/2\\ -1 & when\ \ a/2<t<a\end{cases}$. $Ans:\ \frac{1}{s}\tanh\left(\frac{as}{4}\right)$

2. Find the Laplace transform of the saw-toothed wave of period T, given $f(t)=\frac{t}{T}\ \ for\ \ 0<t<T$. $Ans:\ \frac{1}{s^2T}-e^{-sT}\frac{(1-e^{sT})}{s}$

3. Find the Laplace transform of the triangular wave function of period $2a$ given by $f(t)=\begin{cases}t & 0<t<a\\ 2a-t & a<t<2a\end{cases}$. $Ans:\ \frac{1}{s^2}\tanh\left(\frac{as}{2}\right)$

4. Find the Laplace transform of the full wave rectifier $f(t)=E\sin\omega t$, $0<t<\frac{\pi}{\omega}$ having period $\frac{\pi}{\omega}$. $Ans:\ \frac{E\omega}{s^2+\omega^2}\coth\left(\frac{\pi s}{2\omega}\right)$

5. Find the Laplace transform of the half wave rectifier $f(t)=\begin{cases}\sin\omega t & 0<t<\pi/\omega\\ 0 & \pi/\omega<t<2\pi/\omega\end{cases}$. $Ans:\ \frac{\omega}{\left(1-e^{-\pi s/\omega}\right)\left(s^2+\omega^2\right)}$

6. Find the Laplace transform of the periodic function $f(t)=\begin{cases}1 & 0<t<1\\ 0 & 1<t<2\\ -1 & 2<t<3\end{cases}$ having period 3.

7. Find the Laplace transform of the periodic function $f(t)=\begin{cases}t & 0<t<\pi\\ 0 & \pi<t<2\pi\end{cases}$ having period 2π.

8. Find the Laplace transform of the periodic function $f(t)=\begin{cases}1+t & 0<t\le 1\\ 3-t & 1\le t<2\end{cases}$ having period 2.

9. Find the Laplace transform of the periodic function $f(t)=t^2$, $0<t<\alpha$ and $f(t+\alpha)=f(t)$.

Unit step function or Heaviside function: The unit step function $u(t-a)$ is defined as follows: $u(t-a)=\begin{cases}0 & for\ \ t<a\\ 1 & for\ \ t\ge a\end{cases}$ where a is always positive.

Transform of unit step function: $L\left[u(t-a)\right]=\frac{e^{-as}}{s}$

Second shifting property: If $L\left(f(t)\right)=F(s)$, then $L\left(f(t-a)\cdot u(t-a)\right)=e^{-as}F(s)$ (statement only)

Express the following functions with an interval of the type $(0,\alpha)$ in terms of unit step function. Hence find their Laplace transform.

1. $f(t)=\begin{cases}2t & 0<t<\pi\\ 1 & t>\pi\end{cases}$ $Ans:\ \frac{2}{s^2}+e^{-as}\left(\frac{1-2\pi}{s}-\frac{2}{s^2}\right)$

2. $f(t)=\begin{cases}\cos t & 0<t<\pi \\ \sin t & t>\pi\end{cases}$

3. $f(t)=\begin{cases}e^{-t} & 0<t<3 \\ 0 & t>3\end{cases}$

4. $f(t)=\begin{cases}\sin t & 0<t<\pi \\ \sin 2t & \pi<t<2\pi \\ \sin 3t & t>2\pi\end{cases}$ $Ans:\ \dfrac{1}{s^2+1}+e^{-\pi s}\left(\dfrac{2}{s^2+4}-\dfrac{1}{s^2+1}\right)+e^{-2\pi s}\left(\dfrac{3}{s^2+9}-\dfrac{2}{s^2+4}\right)$

5. $f(t)=\begin{cases}\cos t & for\ \ 0<t\le\pi \\ 1 & for\ \ \pi<t\le 2\pi \\ \sin t & for\ \ t>2\pi\end{cases}$ $Ans:\ \dfrac{s}{s^2+1}+e^{-\pi s}\left(\dfrac{s}{s^2+1}+\dfrac{1}{s}\right)-e^{-2\pi s}\left(\dfrac{1}{s}-\dfrac{1}{s^2+1}\right)$

6. $f(t)=\begin{cases}t^2 & 0<t<2 \\ 4t & 2<t<4 \\ 8 & t>4\end{cases}$ $Ans:\ \dfrac{2}{s^3}\left\{1+e^{-2s}\left(2s^2-1\right)-2e^{-4s}\left(1+4s\right)\right\}$

7. $f(t)=\begin{cases}t^2 & 0<t\le 2 \\ 0 & t>2\end{cases}$ $Ans:\ \dfrac{2(1-e^{-2s})}{s^3}-\dfrac{4e^{-2s}(1+s)}{s^2}$

8. $f(t)=\begin{cases}0 & 0<t<5 \\ t-3 & t>5\end{cases}$

9. $f(t)=\begin{cases}2 & 0<t<\pi \\ 0 & \pi<t<2\pi \\ \sin t & t>2\pi\end{cases}$.

<u>Inverse Laplace Transforms</u>

If $L\left(f(t)\right)=F(s)$ then Inverse Laplace Transform of $F(s)$ is denoted by $L^{-1}\left(F(s)\right)$ and is given as $L^{-1}\left(F(s)\right)=f(t)$.

Inverse Laplace Transforms of Elementary functions

1. $L^{-1}\left(\dfrac{1}{s}\right)=1$, $s>0$
2. $L^{-1}\left(\dfrac{1}{s^{n+1}}\right)=\dfrac{t^n}{n!}$ when n is a positive integer
3. $L^{-1}\left(\dfrac{1}{s-a}\right)=e^{+at}$
4. $L^{-1}\left(\dfrac{1}{s^2+a^2}\right)=\dfrac{\sin at}{a}$
5. $L^{-1}\left(\dfrac{s}{s^2+a^2}\right)=\cos at$
6. $L^{-1}\left(\dfrac{s}{s^2-a^2}\right)=\cosh at$
7. $L^{-1}\left(\dfrac{1}{s^2-a^2}\right)=\dfrac{\sinh at}{a}$

Properties of Inverse Laplace transforms

1. **Linearity property**: If $L^{-1}\left(F(s)\right)=f(t)$, $L^{-1}\left(G(s)\right)=g(t)$ and $L^{-1}\left(H(s)\right)=h(t)$ and let a, b, c be any constants then $L^{-1}\left(aF(s)+bG(s)-cH(s)\right)=af(t)+bg(t)-ch(t)$.
2. **Shifting property**: If $L^{-1}\left(F(s)\right)=f(t)$, then $L^{-1}\left(F(s-a)\right)=e^{+at}f(t)$.
3. **Second Shifting property:** If $L^{-1}\left(F(s)\right)=f(t)$ then $L^{-1}\left(e^{-as}F(s)\right)=f(t-a)\cdot u(t-a)$.

Evaluation of $L^{-1}\left\{F\left(S\right)\right\}$:

1. $\left[\frac{2}{s}-\frac{1}{s^3}\right]^2$ $\qquad Ans: 4t+\frac{t^5}{5!}-4\frac{t^3}{3!}$
2. $\frac{-2s+6}{s^2-4}$ $\qquad Ans: -2\cosh 2t+\frac{3\sinh 2t}{2}$
3. $\frac{(s+1)^3}{s^4}$ $\qquad Ans: 1+\frac{t^3}{6}+3t+\frac{3t^2}{2}$
4. $\frac{4s}{4s^2+1}+\frac{1}{5s^2+1}$ $\qquad Ans: \cos(1/2)t+\sin(1/5)t$
5. $\frac{1}{4s+1}+\frac{1}{5s-2}$ $\qquad Ans: \frac{e^{-(1/4)t}}{4}+\frac{e^{(2/5)t}}{5}$

Evaluation of $L^{-1}\left\{F\left(s-a\right)\right\}$

1. $\dfrac{1}{(s+2)^3}$ $\qquad Ans: \dfrac{e^{-2t}t^2}{2}$
2. $\dfrac{2s-3}{s^2+4s+13}$ $\qquad Ans: 2e^{-2t}\cos 3t - \dfrac{7e^{-2t}\sin 3t}{3}$
3. $\dfrac{s}{s^2+4s+5}$ $\qquad Ans: e^{-2t}\cos t - 2e^{-2t}\sin t$
4. $\dfrac{2s+5}{s^2+6s+34}$ $\qquad Ans: 2e^{-3t}\cos 5t - \dfrac{e^{-3t}\sin 5t}{5}$
5. $\dfrac{s^2}{(s-a)^3}$ $\qquad Ans: \dfrac{a^2e^{at}t^2}{2} + 2ate^{at} + e^{at}$
6. $\dfrac{4}{(s-4)^5}$ $\qquad Ans: \dfrac{4e^{4t}t^4}{4!}$
7. $\dfrac{2s+5}{(s-3)^2}$ $\qquad Ans: 2e^{3t} + 11e^{3t}t$
8. $\dfrac{1}{s^2+2s+5}$ $\qquad Ans: \dfrac{e^{-t}\sin 2t}{2}$
9. $\dfrac{1}{\left(s+4\right)^{5/2}}$ $\qquad Ans: \dfrac{4e^{-4t}t^{(3/2)}}{3\sqrt{\pi}}$

Evaluation of $L^{-1}\left\{F\left(S\right)\right\}$ **using partial fraction:**

1. $\dfrac{s^2+6s+9}{(s-1)(s-2)(s+4)}$ $\qquad Ans: \dfrac{16e^t}{5} - e^{2t} + \dfrac{49e^{-4t}}{30}$
2. $\dfrac{s+1}{s^2-4s}$ $\qquad Ans: \dfrac{-1}{4} + \dfrac{5e^{4t}}{4}$
3. $\dfrac{s^2+1}{s(s-1)(s-2)(s+1)}$ $\qquad Ans: \dfrac{1}{2} - e^t + \dfrac{5e^{2t}}{6} - \dfrac{1}{3}e^{-t}$
4. $\dfrac{\left(s+1\right)^2}{(s+2)^4}$ $\qquad Ans: \dfrac{-e^{-2t}t^3}{6} + e^{-2t}t^2 + e^{-2t}t$
5. $\dfrac{s}{(s+2)(s^2+4)}$
6. $\dfrac{s}{(s-2)(s-3)(s-6)}$
7. $\dfrac{2s-1}{s^2(s+1)^3}$

Evaluation of $L^{-1}\left\{e^{-as}F\left(s\right)\right\}$**:**

1. $\dfrac{e^{-3s}}{\left(s-4\right)^2}$ $\qquad Ans: \left(e^{4(t-3)}\left(t-3\right)\right)u(t-3)$
2. $\dfrac{s\,e^{-\pi s}}{s^2+9}$ $\qquad Ans: -\cos 3t\ u(t-\pi)$

3. $\dfrac{s\,e^{-s/2}+\pi e^{-s}}{s^2+\pi^2}$ $\quad Ans: \sin \pi t\ u\left[t-\dfrac{1}{2}\right]-\sin \pi t\, u(t-1)$

4. $\dfrac{s\,e^{-as}}{s^2-k^2}$ $\quad Ans: \cosh k\left(t-a\right)u(t-a)$

5. $\dfrac{e^{-3s}}{\left(s+4\right)^{5/2}}$ $\quad Ans: \dfrac{4e^{-4(t-3)}\left(t-3\right)^{(3/2)}}{3\sqrt{\pi}}u(t-3)$

6. $\dfrac{\cosh 2s}{e^{3s}s^2}$ $\quad Ans: \dfrac{\left(t-1\right)u(t-1)+\left(t-3\right)u(t-3)}{2}$

7. $\dfrac{e^{-2s}}{s\left(s^2-1\right)}$ $\quad Ans: u(t-2)(\cosh\left(t-2\right)-1)$

8. $\dfrac{e^{-\pi s}}{s^2+1}$ $\quad Ans: -\sin t\ u(t-\pi)$

Application to Differential equations

Solve the following DE by the method of Laplace transform.

1. $y'''+2y''-y'-2y=0$ given $y\left(0\right)=y'\left(0\right)=0$ and $y''\left(0\right)=6$.
2. $\left(D^3-3D^2+3D-1\right)y=t^2e^t$ given $y\left(0\right)=1$, $y'\left(0\right)=0$ and $y'''\left(0\right)=1$.
3. $(D^2+1)y=f(t)$ where f(t)= $\begin{cases}0 & 0\le t<\pi\\ 1 & \pi\le t<2\pi\\ 0 & t\ge 2\pi\end{cases}$ $y\left(0\right)=0$ $y'\left(0\right)=1$
4. $\dfrac{d^2x}{dt^2}+9x=\cos\left(2t\right)$ if $x\left(0\right)=1$ and $x\left(\pi/2\right)=-1$.
5. $\dfrac{d^2y}{dx^2}+y=0$ if $y\left(0\right)=2$ and $y\left(\pi/2\right)=1$.
6. $\left(D^2-1\right)x=a\cosh t$ given $x\left(0\right)=0=x'\left(0\right)$.
7. $\left(D^2-D\right)t=e^t\cos t$ $y\left(0\right)=0$ $y'\left(0\right)=0$
8. $\dfrac{d^2x}{dt^2}-2\dfrac{dx}{dt}+x=e^t$ with $x=2$, $\dfrac{dx}{dt}=-1$, at $t=0$.
9. In a single loop LR circuit the current I builds up at the rate given by $L\dfrac{dI}{dt}+RI=E(t)$.

   Determine the current I(t), when L=1H, R=1Ω, and $E(t)=\begin{cases}t & 0<t<1\\ 0 & t>1\end{cases}$ subject to the condition $\mathrm{I}(0)=0$.
10. In an electrical circuit with e.m.f E(t), inductance L ,resistance R and capacitance C the current i builds up at the rate given by $L\dfrac{di}{dt}+Ri+\dfrac{1}{C}\int_0^t i(t)dt=E(t)$. Determine the current i(t) when L=0.1H, R=3Ω, C=0.05F and $E(t)=100[u(t-1)-u(t-2)]$ with the zero initial condition .

    $Ans: i(t)=100\left[e^{-10(t-1)}-e^{-20(t-1)}\right]u(t-1)-100\left[e^{-10(t-2)}-e^{-20(t-2)}\right]u(t-2)$